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# Simulation of the Effectiveness Assessment Model for University Student Psychological Counseling Data Based on Deep Learning Algorithms

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***Abstract*—Psychological counseling for university students has received much attention in recent years, and how to effectively assess the effectiveness of a large amount of counseling data has become an urgent problem to be solved. This article aims to propose a validity assessment model for university student psychological counseling data based on deep learning (DL), and compare it with traditional decision tree (DT) algorithm to verify its superiority and practicality. Based on the characteristics of the data and the actual needs of assessment, this study will carefully design the network structure and optimize the model through training to ensure that it can accurately capture effective features from the data. Through comparative experiments, the DL algorithm proposed in this article is significantly superior to the DT algorithm in terms of prediction accuracy, operational efficiency, and model stability. The DL algorithm has shown significant advantages in evaluating the effectiveness of psychological counseling data for university students. This algorithm can more accurately assess the effectiveness of counseling data, provide strong support for psychological counseling services for university students, and promote progress and development in this field.**

***Keywords-Deep learning; University students; Psychological counseling data; Effectiveness assessment***

## I. Introduction

In today's fast-paced society, the psychological well-being issues of university students are increasingly prominent and have become a focus of attention for education and society[1]. In order to address this issue, psychological counseling services for university students are gradually receiving attention, and the assessment of the effectiveness of psychological counseling data has become a core link in improving service quality. Traditional assessment methods are usually based on manual experience and questionnaire surveys, but these methods have certain subjectivity and limitations, and cannot fully explore the deep information behind the data [2]. Therefore, how to accurately and objectively assess the effectiveness of psychological counseling data has become an urgent problem to be solved. Based on the progress of DL algorithm, it has achieved significant results in fields such as data mining and pattern recognition. DL has strong feature extraction and classification capabilities, and can automatically learn high-level abstract features in data, making it suitable for processing and analyzing complex data [3-4]. This provides a new solution for evaluating the effectiveness of psychological counseling data for university students.

This article constructs a validity assessment model for university student psychological counseling data based on DL algorithm and conducts simulation experiments on it. Firstly, preprocess the collected psychological counseling data of university students and extract key features. Then, use the DL algorithm to construct an assessment model, learn potential patterns in the data, and achieve effectiveness assessment. Finally, the performance of the assessment model is verified through simulation experiments, and the experimental results are analyzed. The research results will provide a new and objective method for evaluating the effectiveness of psychological counseling data for university students, overcoming the limitations of traditional methods. Moreover, the research in this article also contributes to improving the quality and efficiency of psychological counseling services for university students, providing strong support for ensuring their psychological well-being.

## II. Theoretical basis

The theoretical framework behind DL is based on neural network model, which skillfully combines low-level features to generate more abstract high-level representation attribute categories or features. This feature enables DL to discover the distributed feature representation of data and further reveal the deep structure and laws behind the data.

Traditional assessment methods are limited by manual experience and questionnaire survey, while DL algorithm overcomes these limitations. By customizing the DL model for this task, we can automatically and efficiently extract the key features from the psychological counseling data, and at the same time learn the hidden patterns behind the data [5]. This method not only improves the accuracy and objectivity of the assessment, but also greatly reduces human intervention and improves the assessment efficiency. In order to achieve this goal, this paper will deeply study and select the appropriate DL algorithm, and further customize the validity assessment model of university students' psychological counseling data. According to the characteristics of data and the

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actual needs of assessment, this study will carefully design the network structure and train the optimization model to ensure that it can accurately capture the effectiveness characteristics from the data. Through the theory and application of DL, it is expected to bring a more scientific and efficient method for the validity assessment of university students' psychological counseling data, and provide strong support for the subsequent psychological counseling services, and help university students' psychological healthy growth.

## III. Validity Assessment Model of University Students' Psychological Counseling Data

As a powerful machine learning technology, DL has been widely used in the processing and analysis of various complex data. DL provides an automatic and accurate method to assess the validity of university students' psychological counseling data. When constructing the assessment model, we need to make clear the design idea of the model first. For psychological counseling data, its validity may involve many aspects, such as the depth of counseling content, the response style of counselors, and the feedback from students [6]. Therefore, the DL model designed in this paper needs to be able to automatically capture these various information and give a comprehensive assessment of its effectiveness.

A large quantity of psychological counseling data of university students were collected in the study. These data come from university psychological counseling centers and online psychological counseling platforms. The forms of data include text recording, audio, video, etc. When collecting data, it is needed to ensure the privacy and compliance of the data and abide by relevant ethical regulations. After the data is collected, the data is preprocessed to prepare for the subsequent data mining work. The purpose of data preprocessing is to clean data, eliminate noise, and deal with missing values [7]. Specific preprocessing steps can include word segmentation, stop words removal, stem extraction, data standardization and so on [8]. These steps help to transform the data into a form suitable for the DL model. In the process of feature extraction and selection, it is needed to consider the relevance and importance of extracting the features related to the effectiveness assessment of psychological counseling data from the preprocessed data, and can be screened by means of feature importance analysis and correlation analysis. The data mining process of the validity assessment model of university students' psychological counseling data is shown in Figure 1.

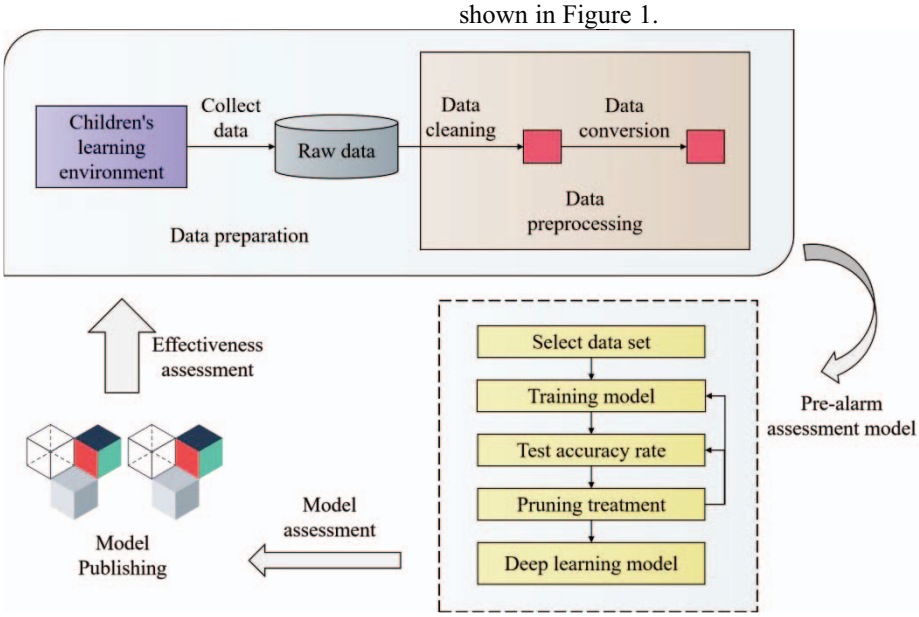


Figure 1. Data mining process of assessment model

The input layer of the model directly receives the preprocessed psychological counseling data. These data can be in the form of text, audio, video, etc., and after being properly encoded, they are sent to DL model. The feature extraction layer is mainly used to automatically extract key features from input data. The context fusion layer fuses the features of different time steps to capture the global context information [9]. This helps the model to understand the overall situation in the consultation process, so as to better assess the validity of the data. The output layer finally gives the assessment result of data validity. When training the model, supervised learning is adopted. Using the labeled data set, the parameters of the model are constantly adjusted through back propagation and optimization algorithm, so that the gap between the predicted results of the model and the real labeling is gradually reduced. In order to enhance the generalization ability of the model, regularization, dropout and other technologies will be used to prevent over-fitting.



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The weight of each characteristic index is given by experts according to their experience, and then the average value of the weight given by experts is taken as the weight of each characteristic index:

$$\eta = \{\eta_1, \eta_2, \cdots, \eta_n\} \quad (1)$$

$\eta_j$ reflects the position or role of the $j$ index in all characteristic indexes, which directly affects the results of cluster analysis.

Calculate the comprehensive clustering coefficient $\sigma_i^k$ of object $i$ about grey class $k$ as follows:

$$\sigma_i^k = \sum_{j=1}^{m} f^k(y_{ij}) * \eta_j \quad (2)$$

$f^k(y_{ij})$ is the whitening weight function of $k$ subclass, and $\eta_j$ is the weight of $j$ index.

Establish several data subsets of size $n$, and set the sample weight of each subset to 1. By calculating the gray clustering center $y$ of the first group of samples with a weight of 1, the smallest sum of squares of clustering errors $D(y)$ is obtained:

$$D(y) = \min_{1 \le k \le n-1} \sum_{i=1}^{n} \omega_i (y_i - y_{i,k})^2 \quad (3)$$

Where: $\omega_i$ is the weight of the $i$ th index; $y_i$ is the database mapping value of the $i$ th indicator; $y_{i,k}$ is the mapping value of the $i$ th index database of the $k$ th cluster center.

After feature extraction and selection, these features can be used to train DL model. By choosing the appropriate DL algorithm and designing the appropriate network structure, we can construct the validity assessment model of university students' psychological counseling data [10]. In the training process, the labeled data sets are used for supervised learning, and the model parameters are optimized by back propagation algorithm to achieve more accurate prediction. Determine the index universe $A$ for the assessment object;

$$A = \{a_1, a_2, a_3, \ldots, a_n\} \quad (4)$$

Among them, $n$ represents the quantity of assessment indicators, and the assessment system of university students' psychological counseling data adopts a two-layer structure, with the assessment object being the effectiveness of university students' psychological counseling data, and the index universe representing the first-class indicators. Determine the assessment level domain $B$:

$$B = \{b_1, b_2, b_3, \ldots, b_m\} \quad (5)$$

Among them, $m$ represents the quantity of assessment grades, and each grade has a one-to-one correspondence with each fuzzy subset. Generally, $m$ is odd, so it is convenient to use its middle grade for assessment. Quantify all assessment indexes in turn for the items to be assessed, and the fuzzy relation matrix $C$ is determined according to the membership degree $C(a_i)$ of each level of the fuzzy subset:

$$C = \begin{bmatrix} C(a_1) \\ C(a_2) \\ \cdots \\ C(a_n) \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ & \cdots & \cdots & \\ c_{n1} & c_{n2} & \cdots & c_{nm} \end{bmatrix} \quad (6)$$

Where $C$ is the fuzzy relationship from $A$ to $B$, and $c_{ij}$ represents the first $j$ assessment result obtained by the first $i$ factor.

Through the above model design, training, and optimization, a DL based model for evaluating the effectiveness of university student psychological counseling data can be established. This model can automatically and accurately assess the effectiveness of psychological counseling data, providing strong support for psychological counseling services for university students.

## IV. Result Analysis and Discussion

The purpose of the experiment is to verify the performance of the DL based model for evaluating the effectiveness of university student psychological counseling data, and to compare it with the DT algorithm to demonstrate its superiority and practicality. The psychological counseling data of university students used in the experiment was collected through professional channels, ensuring the privacy and compliance of the data. These data include text, audio, and other forms, covering various psychological counseling scenarios for university students. After preprocessing, the data is divided into training and testing sets for model training and validation. Design and construct an assessment model for the effectiveness of psychological counseling data for university students based on DL technology. The model includes an input layer, a feature extraction layer, a context fusion layer, and an output layer. Select the DT



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algorithm as the comparison algorithm for performance comparison with the DL algorithm proposed in this article. By comparing Figure 2, Figure 3, and Figure 4, it is clear that different models have different performance differences in evaluating the effectiveness of university student psychological counseling data.

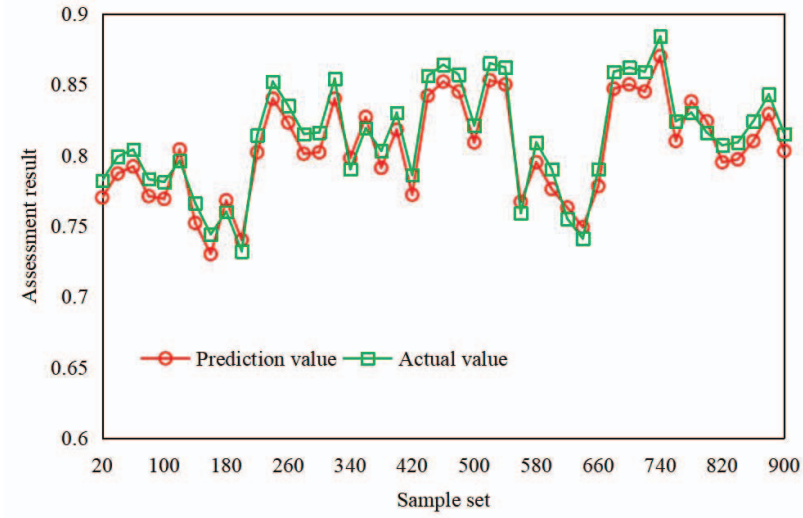


Figure 2. Comparison between predicted value and actual value of this model

Figure 2 shows the comparison between the predicted value and the actual value based on DL model proposed in this paper. As can be seen from the figure, the distribution of the predicted value is very close to the actual value, and most of the predicted points are located near the actual value, which shows that the model has high prediction accuracy. DL model can more accurately assess the validity of university students' psychological counseling data by automatically extracting key features from the data and learning complex data patterns.

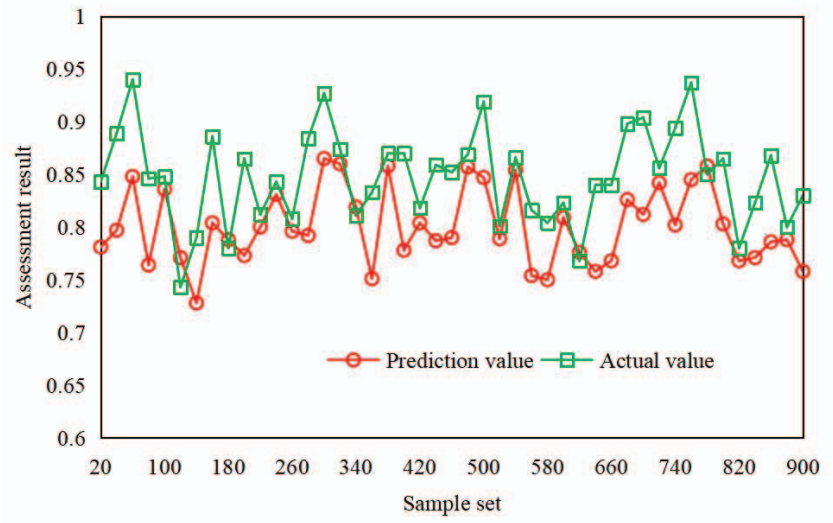


Figure 3. Comparison between predicted value and actual value of DT algorithm

Figure 3 shows the comparison between the predicted value and the actual value of DT algorithm. In contrast, the predicted value of DT algorithm is in low agreement with the actual value, and there are some obvious deviations. This may be because DT algorithm is limited by its tree structure and splitting rules when dealing with complex and nonlinear psychological counseling data, and it can't fully capture the deep features and patterns in the data, which leads to the decline of prediction accuracy.



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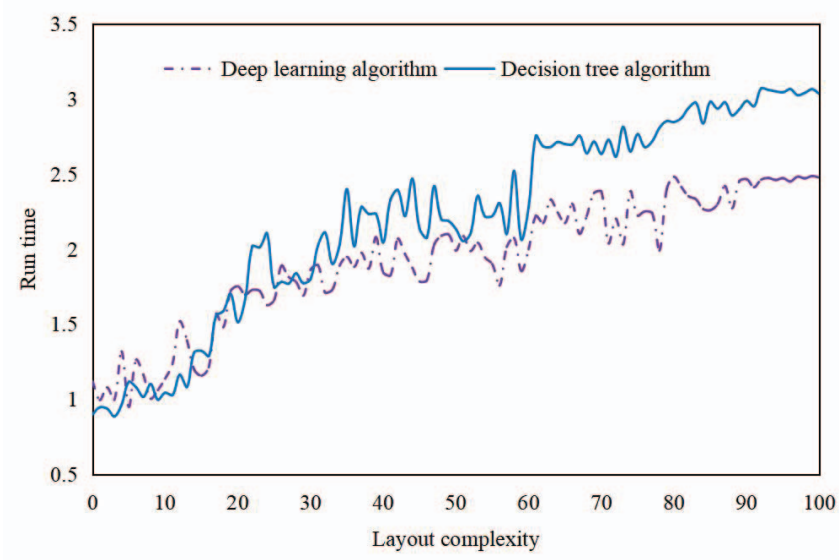


Figure 4. Comparison of algorithm running time

Figure 4 shows the comparison of running time between DL algorithm and DT algorithm. It can be clearly seen from the figure that the running time of DL algorithm proposed in this paper is significantly shorter than that of DT algorithm. This is mainly due to the computational parallelism of DL algorithm and GPU accelerated optimization, which enables it to process large-scale data more efficiently. However, DT algorithm needs many recursions and feature splitting in the running process, which leads to high time complexity.

The error test results of different algorithms in evaluating the validity of psychological counseling data reveal the performance difference between this algorithm and DT algorithm. Compared with DT algorithm, this algorithm shows significant advantages in the later stage, and the error rate decreases by 28.99%.

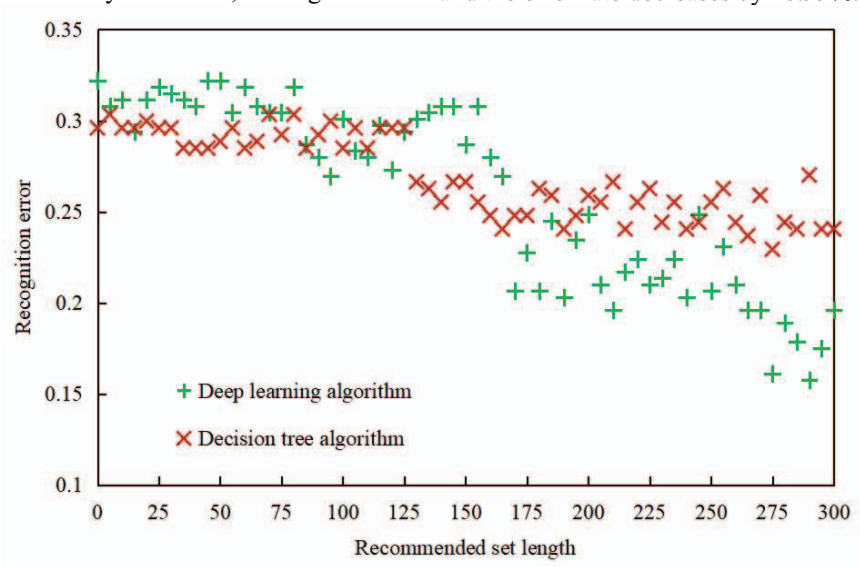


Figure 4. Assessment error

DL algorithm has powerful automatic feature learning ability. Through the multi-layer neural network structure, this algorithm can capture more complex characteristics of psychological counseling data, and understand the deep pattern behind the data, so as to assess the effectiveness more accurately. Although DT performs well in many tasks, its simple model based on tree structure may be difficult to fully capture the complex patterns and nonlinear relationships in psychological counseling data. DT is relatively sensitive to the noise and abnormal values of data, and there may be various uncertainties in psychological counseling data, which makes DT algorithm may produce higher error rate in assessment. By comparing and analyzing the error test



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results, we can clearly see the advantages of the DL-based algorithm proposed in this paper in evaluating the validity of psychological counseling data. This advantage is not only reflected in higher assessment accuracy, but more importantly, it introduces a more effective and reliable assessment tool for the field of psychological counseling.

By comparing the prediction results and actual values of this algorithm with DT algorithm, it is obvious that DL algorithm has higher prediction accuracy. This means that in the face of complex and diverse university students' psychological counseling data, DL algorithm can better capture the key information in the data, and then provide more accurate data validity assessment for psychological counselors. By accurately evaluating the validity of psychological counseling data, psychological counselors can quickly identify the counseling contents and strategies that are helpful to students, so as to adjust the counseling methods and improve the counseling effect. Efficient data validity assessment means that counselors can get valuable feedback faster, and then reduce ineffective or inefficient counseling strategies. Accurate data validity assessment can also provide reliable research tools and data support for psychological counseling researchers and promote further development and innovation in this field.

## V. Conclusions

The psychological well-being problems of university students are increasingly prominent, which has become the focus of education and social attention. In order to deal with this problem, university students' psychological counseling service has been paid more and more attention. How to assess the validity of psychological counseling data accurately and objectively has become an urgent problem to be solved. DL has strong ability of feature extraction and classification, can automatically learn high-level abstract features in data, and is suitable for the processing and analysis of complex data. Through in-depth research and analysis, this paper puts forward a DL-based assessment model for the validity of university students' psychological counseling data. This model is obviously superior to DT algorithm in many aspects. From prediction accuracy to operation efficiency, and then to the stability of the model, DL algorithm has shown its strong potential and application value. For university students' psychological counseling, an efficient assessment model undoubtedly provides a powerful tool for psychological counselors. It can help counselors understand students' needs more accurately, improve the consulting effect and provide more targeted help for students. Moreover, this technical progress has also opened up new ideas and directions for psychological counseling research, and promoted the continuous development of academic research and practical application in the field.

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